What's new in .NET MAUI for .NET 11



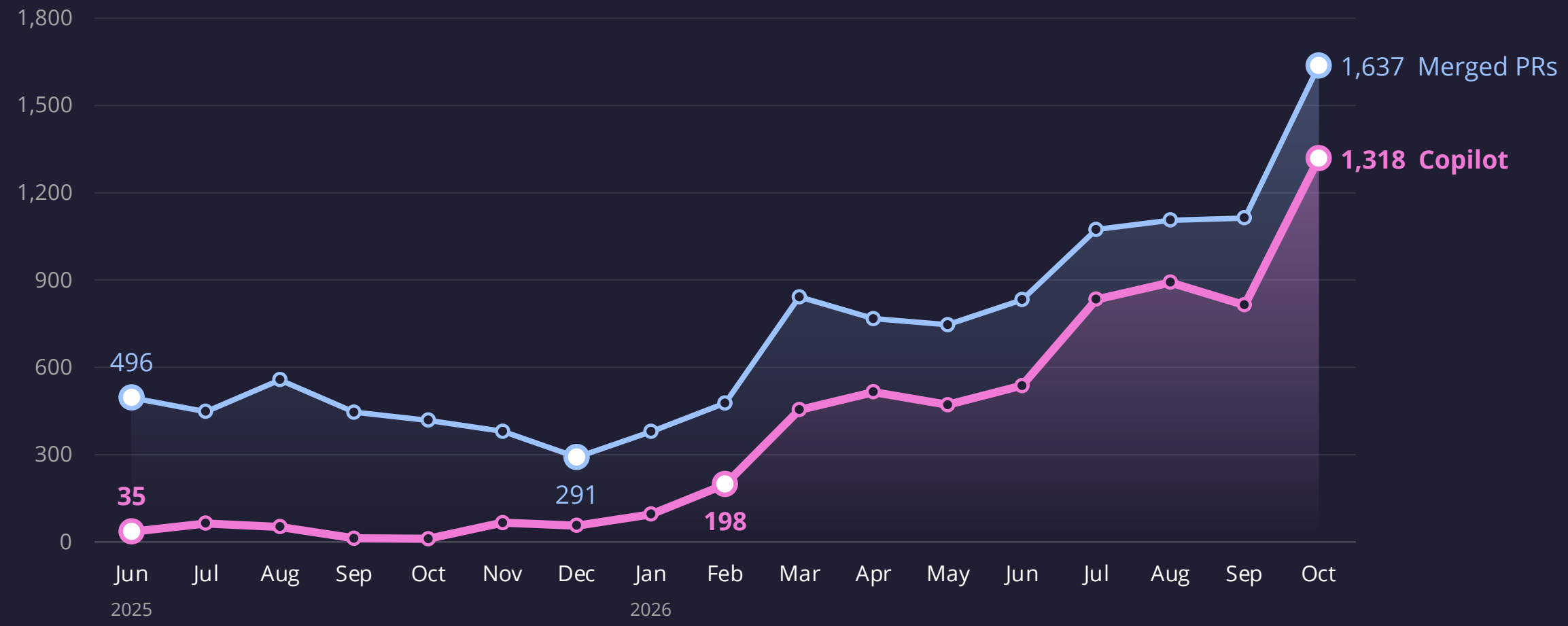
Gerald Versluis

Software Engineer, Microsoft

PR velocity across .NET MAUI repos

Merged PRs grew 4x in a year, and 80% are now Copilot-authored

Monthly merged PRs, .NET MAUI repos



Source: Agentic Engineers PR dashboard, https://agentic-engineers.net, data as of Nov 5, 2026. Copilot = PRs authored with GitHub Copilot.

.NET MAUI ecosystem

Partners are building the framework with us



CONTROLS AND FIXES

Syncfusion

- Engineers fixing issues and shipping features in dotnet/maui
- Free, open-source Syncfusion Toolkit for .NET MAUI



PLATFORM AND RUNTIME

Uno Platform

- Helped deliver Android support in .NET 10 & .NET 11
- Co-maintains SkiaSharp
- Contributes to and reviews PRs on .NET for Android and .NET for iOS



NEW PLATFORMS

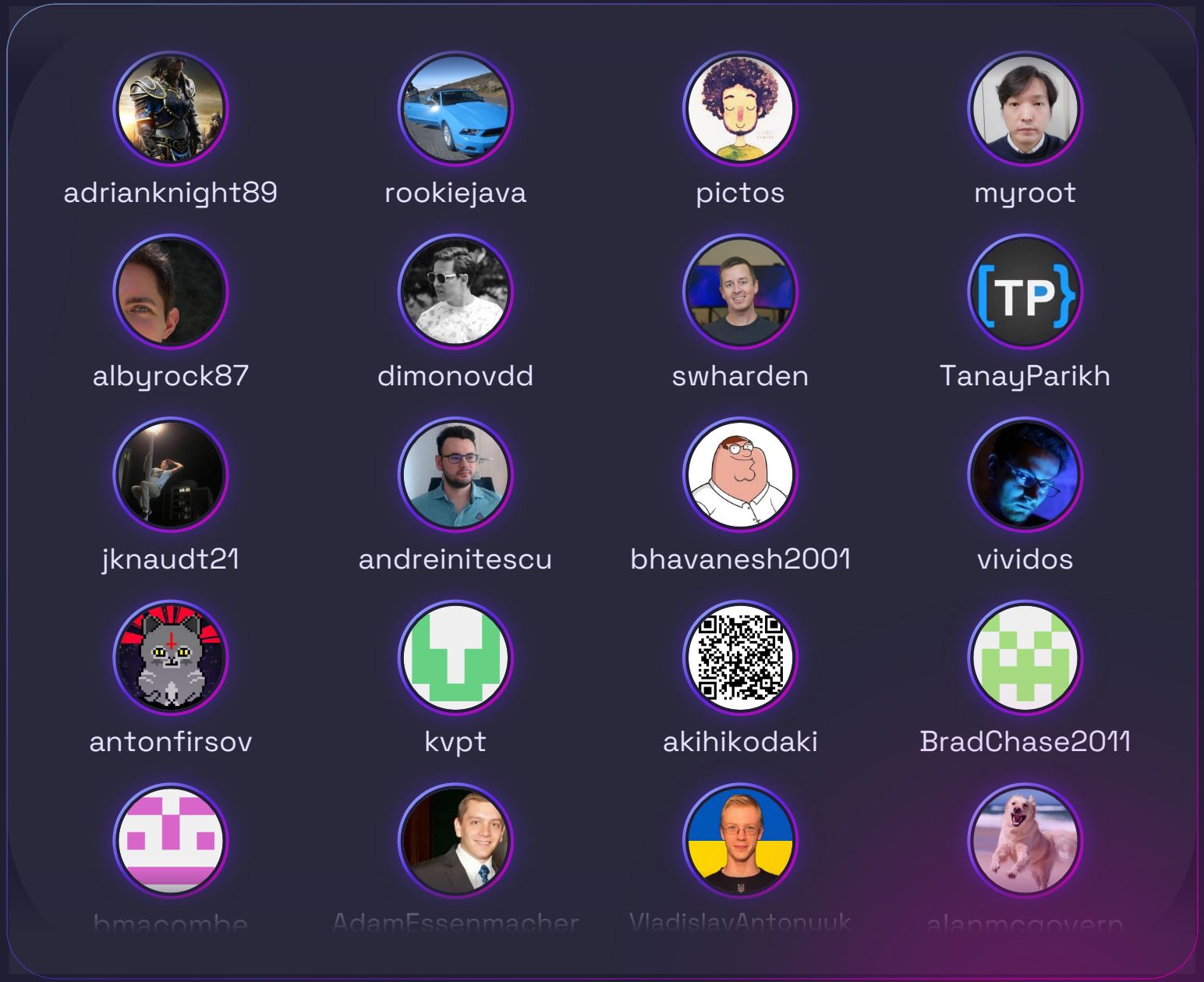
Avalonia

- Building the backend that takes .NET MAUI to Linux and WebAssembly
- In preview since March 2026
- Now an option in the .NET MAUI project templates

github.com/dotnet/maui

Thank you, community contributors!

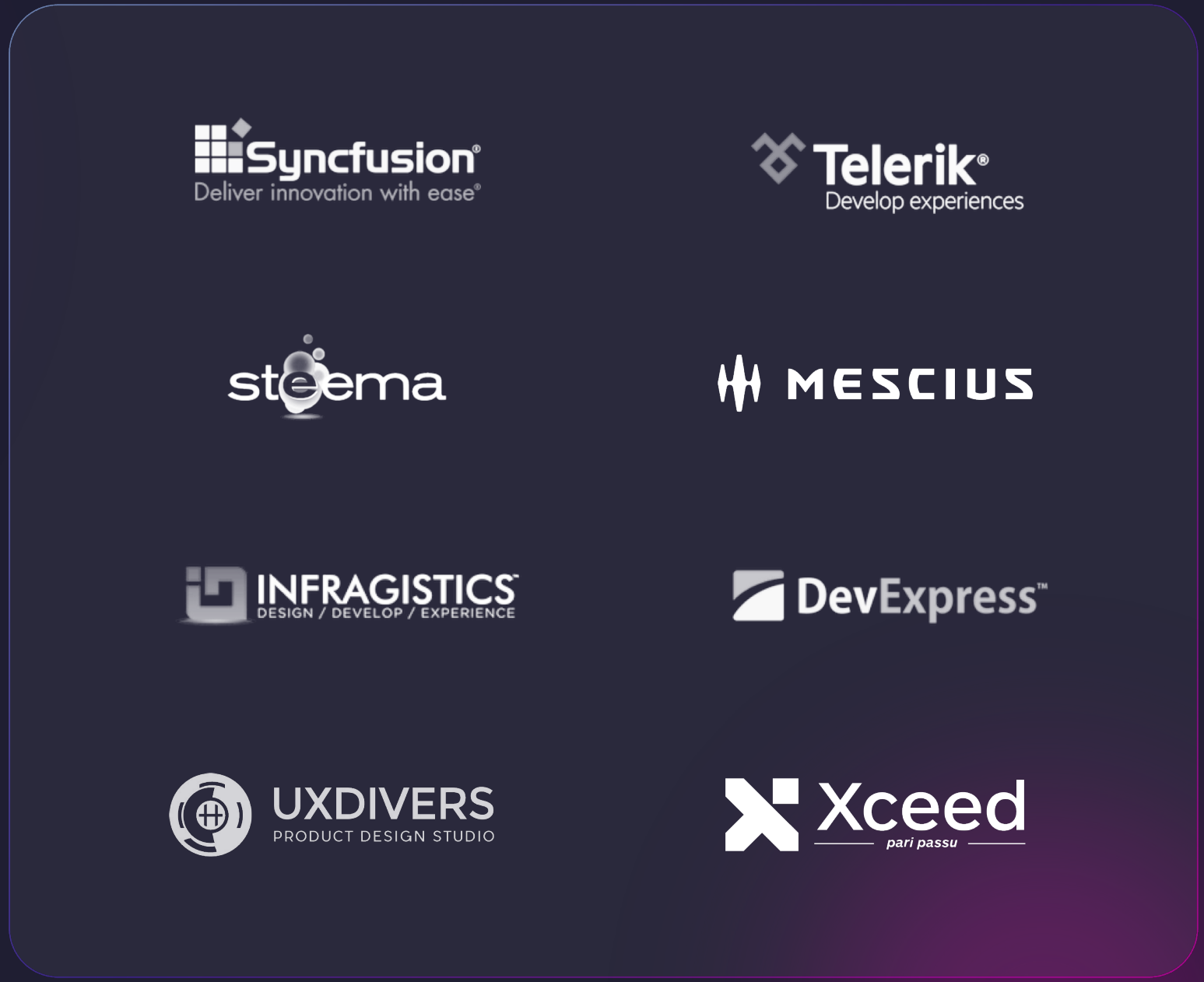
Every pull request, issue and review makes .NET MAUI better for everyone.



.NET MAUI ecosystem

Thank you, component vendors!

Their control suites help .NET MAUI developers ship polished apps faster.

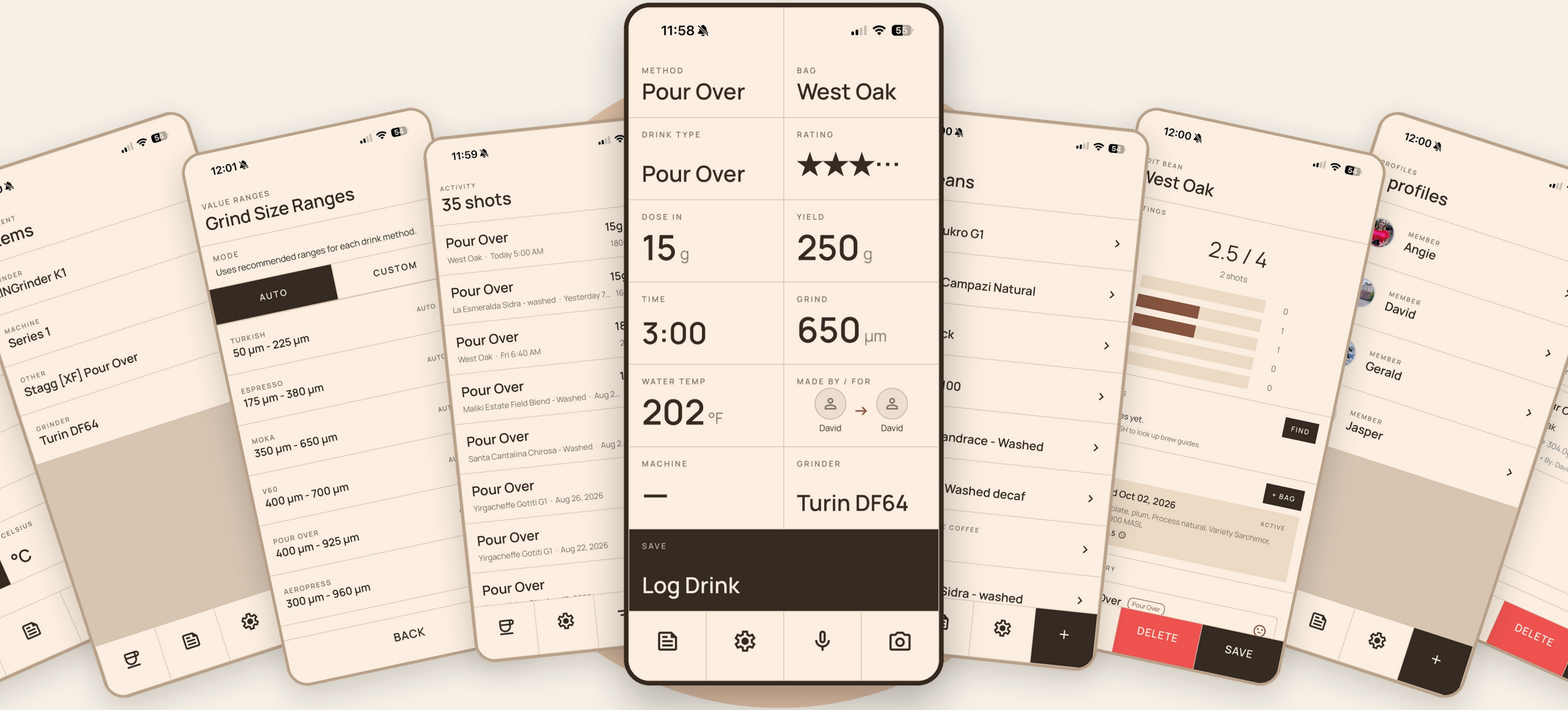


01 CoreCLR



The biggest change to .NET MAUI you won't see, but you will feel.

Barista Notes

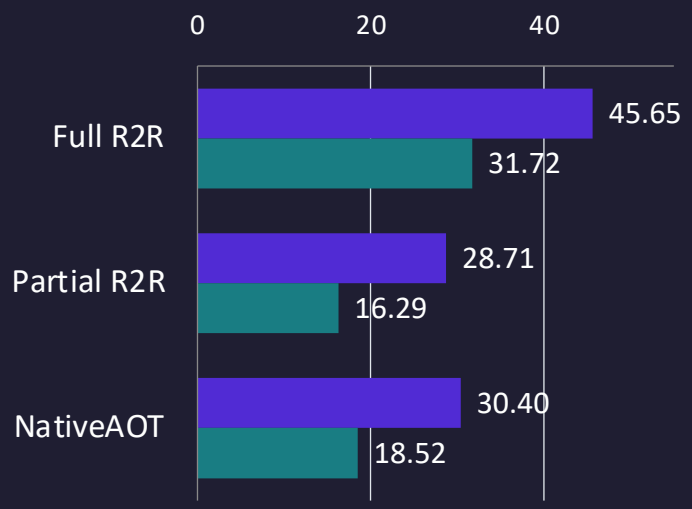


BaristaNotes: Android package size and startup

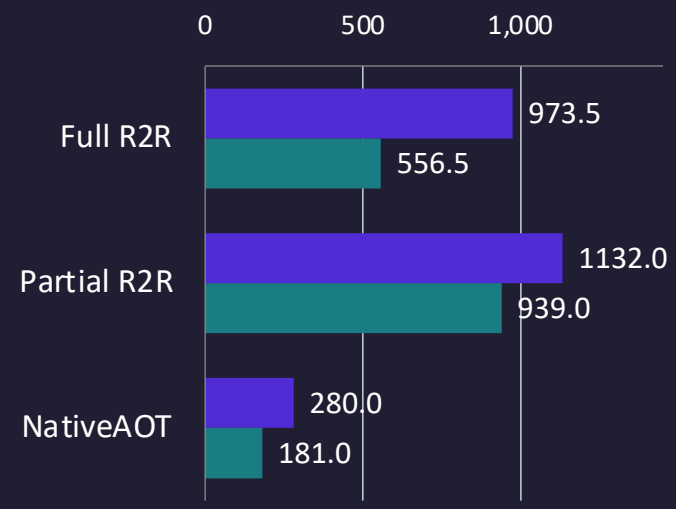
Pixel 5, Android 14 | .NET 11 RC2 | 1,000 drinks | 30 launches per cell

MAUI C# Android Views

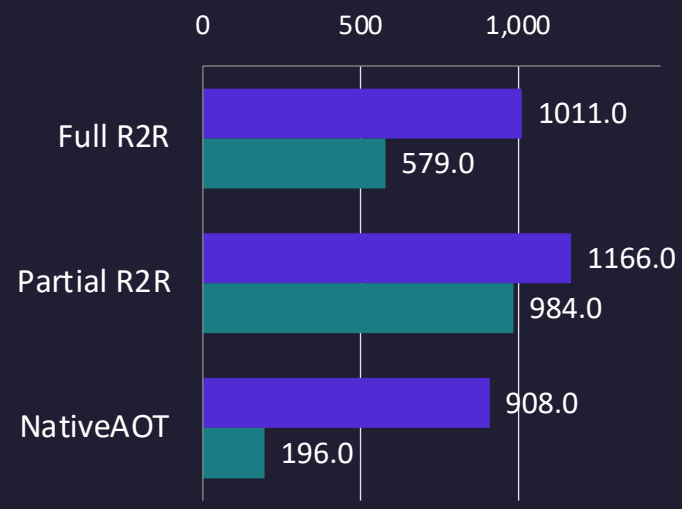
Signed package (MiB)



Median startup (ms)



90th percentile (ms)

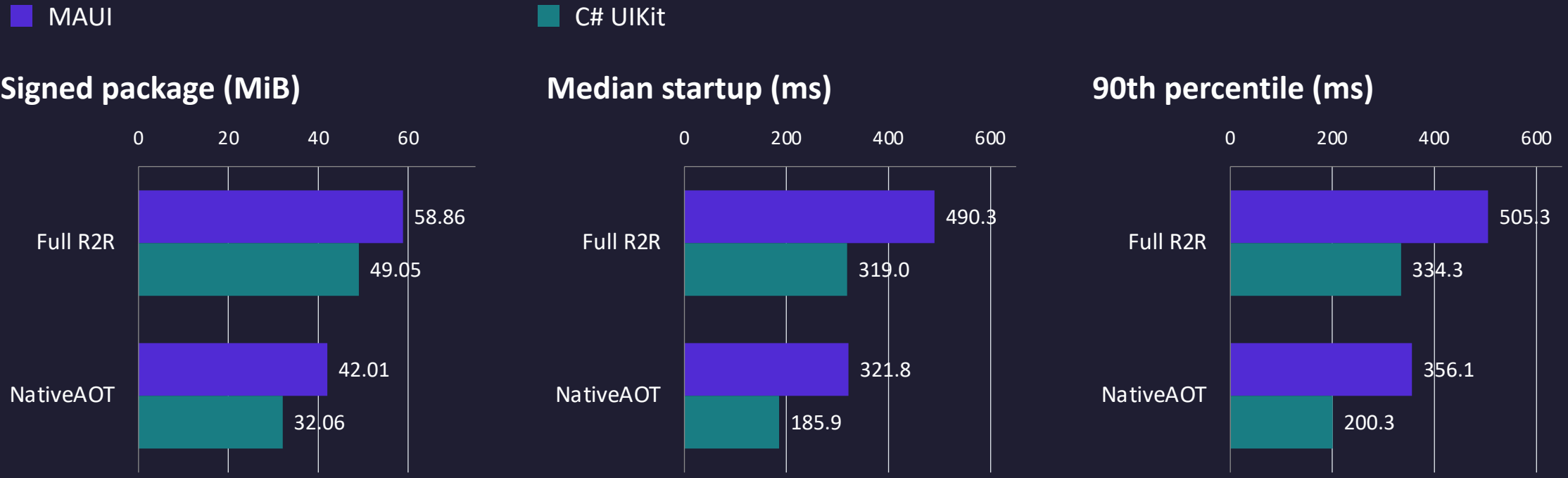


Each panel has its own scale; zero baseline throughout.

Process-cold startup uses Android TotalTime. Signed benchmark packages include the fixture and probes. Android NativeAOT is experimental.

BaristaNotes: iOS package size and startup

iPhone 15 Pro, iOS 26.7.1 | .NET 11 RC2 | 1,000 drinks | 30 launches per cell



Each panel has its own scale; zero baseline throughout.

Startup: kernel process start to New Drink readiness plus two frames. MAUI NativeAOT has an accepted compiler warning; measured routes only.

02 XAML improvements



20 years of XAML, from WPF to .NET MAUI. Familiar syntax, brand-new tricks.

XAML source generation

Every XAML file now compiles to C# at build time

```
// Nothing to add in .NET 11
// MainPage.xaml → MainPage.xaml.sg.cs

<!— Need the old pipeline? Opt out —>
<PropertyGroup>
  <MauiXamlInflator>XamlC</MauiXamlInflator>
</PropertyGroup>

<!—  ... or only for some files —>
<MauiXaml Update="Legacy/*.xaml"
          Inflator="XamlC" />
```

How XAML gets inflated, by default

	.NET 10	.NET 11
Debug	Runtime	**SourceGen**
Release	XamlC	**SourceGen**

Faster builds and startup
No XAML parsing at runtime, in any configuration

Debug behaves like Release
One pipeline from F5 to the store build

Errors at build time
Not when the page first loads

Less XAML ceremony

No more xmlns boilerplate on every page

Before: every page declares its namespaces

```
<ContentPage
  xmlns="http://schemas.microsoft.com/dotnet/2021/maui"
  xmlns:x="http://schemas.microsoft.com/winfx/2009/xaml"
  xmlns:vm="clr-namespace:MyApp.ViewModels"
  xmlns:conv="clr-namespace:MyApp.Converters"
  x:Class="MyApp.MainPage">
  <vm:MainViewModel />
</ContentPage>
```

.NET 11

```
<ContentPage
  x:Class="MyApp.MainPage">
  <MainViewModel />
</ContentPage>
```

Map your own namespaces once per assembly: `[assembly: XmlnsDefinition( ... )]`

C# expressions in XAML

Inline C# replaces most value converters

Before: a converter for one inverted bool

```
// 1. Write a converter class
public class InverseBoolConverter : IValueConverter
{
  public object Convert(object v, Type t,
      object p, CultureInfo c) => !(bool)v;
  public object ConvertBack(object v, Type t,
      object p, CultureInfo c) => !(bool)v;
}
// 2. Register it as a resource
<local:InverseBoolConverter x:Key="InverseBool" />
// 3. Reference it in the binding
<Button IsEnabled="{Binding IsBusy,
   Converter={StaticResource InverseBool}}" />
```

.NET 11

```
<Button
  IsEnabled="{!IsBusy}" />

// Same idea, other converters
{$'Hello {Name}!'}
{Price * Quantity}
{Title ?? 'Untitled'}
{IsAdmin || IsOwner}
```

03 New features & bugfixes



A lot more in .NET MAUI 11

Long press gesture
New recognizer for press-and-hold actions

Tab & toolbar badges
BadgeText on tabs and toolbar items

Passkeys
Face ID, Windows Hello and Android biometrics

Material 3
Android controls follow Material 3 design

Maps
Pin clustering, custom pin images, JSON styles

A stronger foundation

Quality, stability & performance across all platforms

Animation control
Cancel one animation at a time

BoxView gradients
Fill accepts any brush, including gradients

C# UI template
Start a new app in C# with --ui csharp

System bar colors
Status and nav bars match your app

Themed splash screens
Separate images for light and dark mode

All changes in the release notes: learn.microsoft.com/dotnet/maui/whats-new/dotnet-11

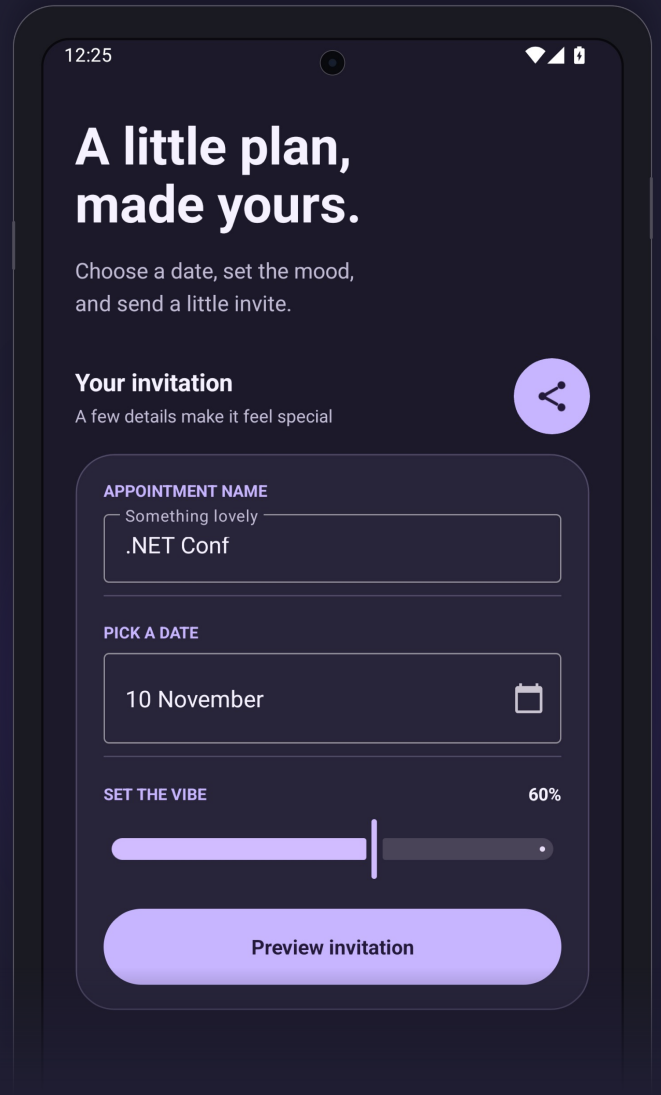
Material 3

One build property modernizes your Android controls

```
<!-- Opt in, in your .csproj -->
<PropertyGroup>
  <UseMaterial3>true</UseMaterial3>
</PropertyGroup>

<!-- Your XAML stays the same -->
<Entry Placeholder="Email" />
<DatePicker />
<Slider Value="0.6" />

// .NET 11: Material 3 types are public
// for custom handlers
class MyEntry : MauiMaterialEditText
```



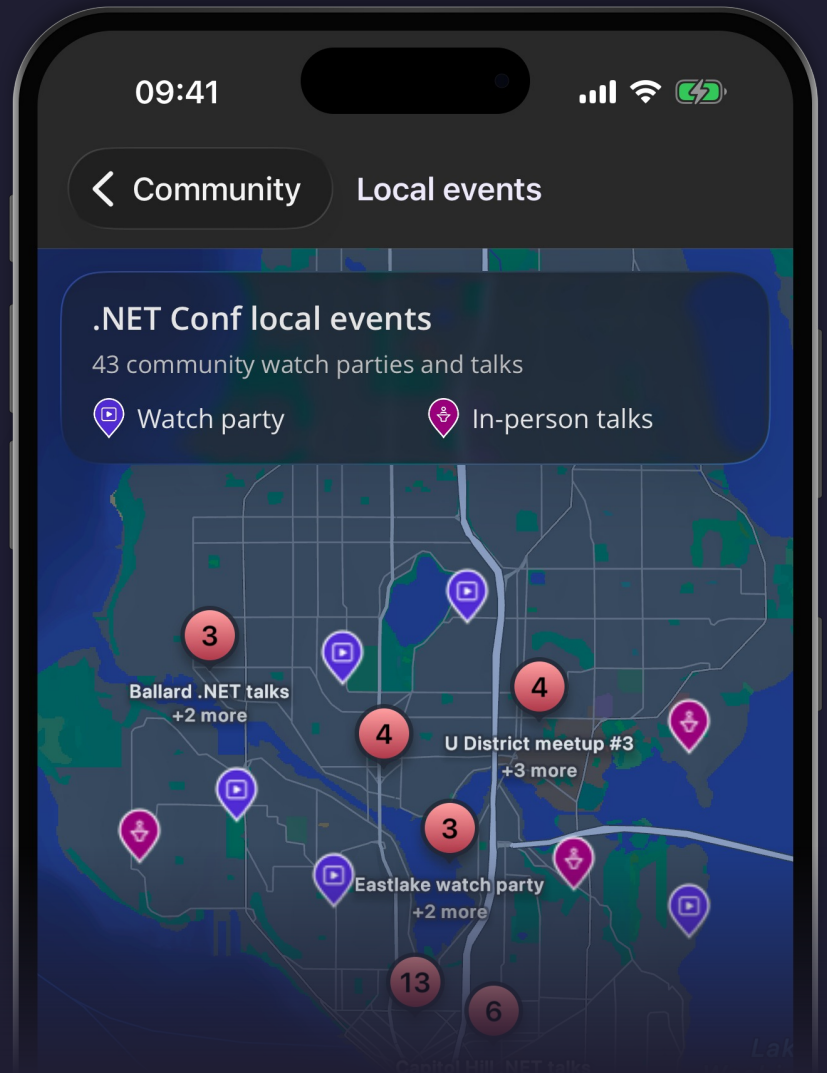
Maps

Hundreds of pins, still one readable map

```
<maps:Map IsClusteringEnabled="True"
          ClusterClicked="OnClusterClicked">

  <maps:Pin Label="Seattle"
            Location="47.6062,-122.3321"
            ClusteringIdentifier="stores"
            ImageSource="store.png" />

</maps:Map>
```



The .NET Conf app, built with .NET MAUI 11, is on the App Store and Google Play
Source: github.com/jfversluis/dotnetconf-app

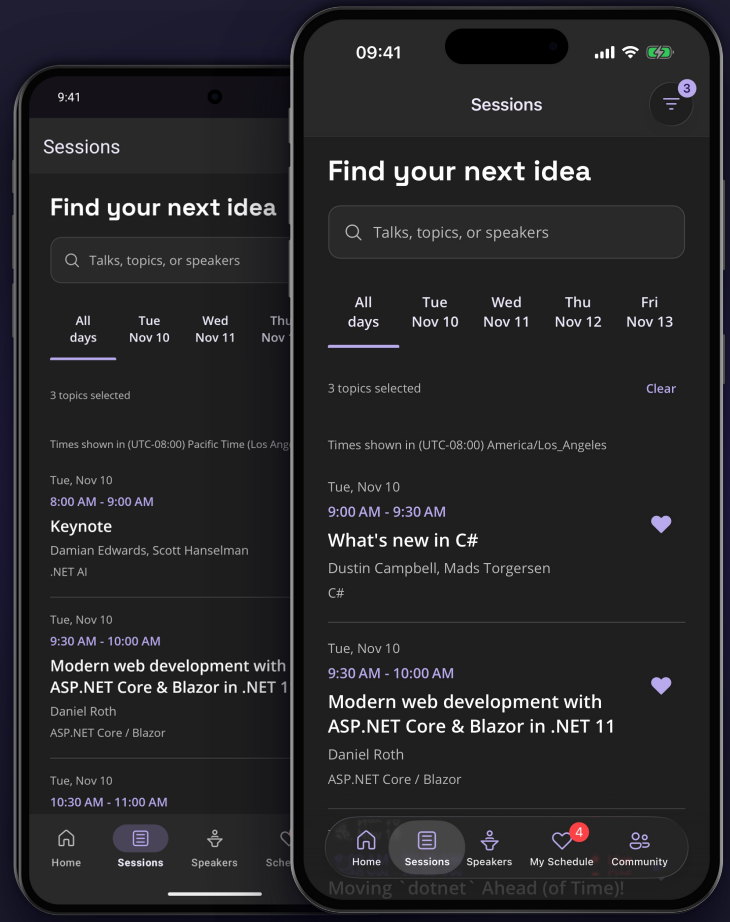
Badges on tabs and toolbar items

Counts and status dots with one property

```
// Toolbar item, bound to the view model
<ToolbarItem Text="Inbox"
  BadgeText="{Binding UnreadCount}" />

// Tab on a TabbedPage
<ContentPage Title="Inbox"
  TabbedPage.BadgeText="3"
  TabbedPage.BadgeColor="Red" />

// "" shows a dot, null hides the badge
```



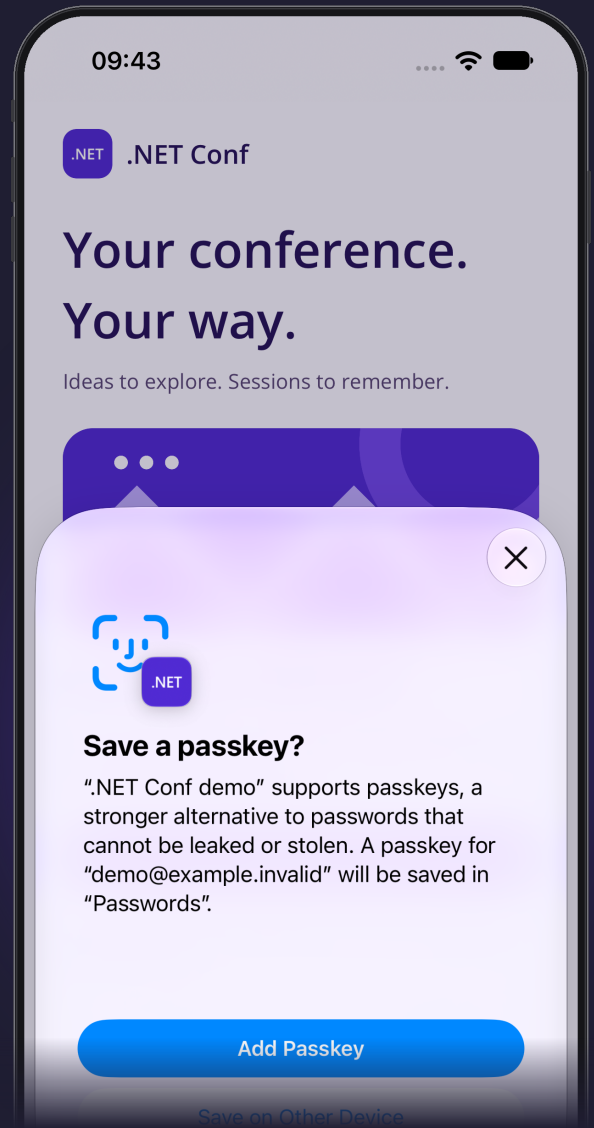
Passkeys

Sign in with Face ID, Windows Hello or a fingerprint

```
using Microsoft.Maui.Authentication;

if (Passkeys.IsSupported)
{
    // Standard WebAuthn JSON in and out
    var response = await Passkeys
        .AssertAsync(optionsJson);

    await server.VerifyAsync(response);
}
```



04 Tooling



dotnet run, test and watch

The .NET CLI now runs your app on devices

dotnet run

```
# Picks the target, then
# the device or emulator
# App output streams here
```

dotnet test

```
# Runs your tests in an
# app on the device
# Results come back here
```

dotnet watch

```
# Hot Reload on Android
# and iOS devices
# Ctrl+R restarts the app
```

DEEP DIVE · BRK016 · Tuesday, 4:30 PM PST

Bringing dotnet run, test, and watch to .NET MAUI with Jonathan Peppers

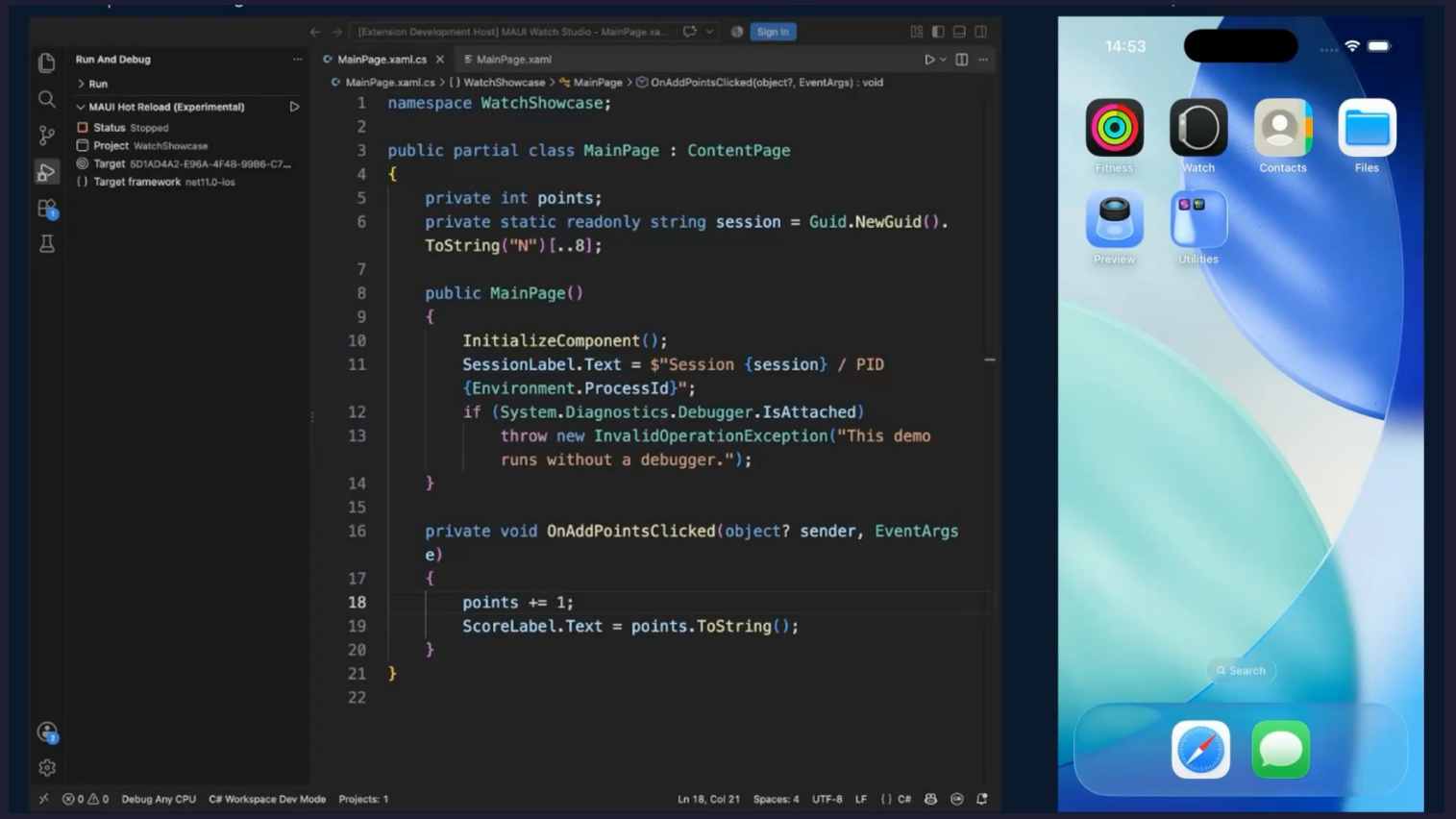
XAML Incremental Hot Reload

Edit XAML, keep your place in the app

```
# On by default in Debug
# Patches live pages:
#   properties, bindings
#   resources
#   add, remove, reorder

# CollectionView keeps
# selection and scroll

# Visual Studio and
# dotnet watch
dotnet watch
```



.NET MAUI Inspector

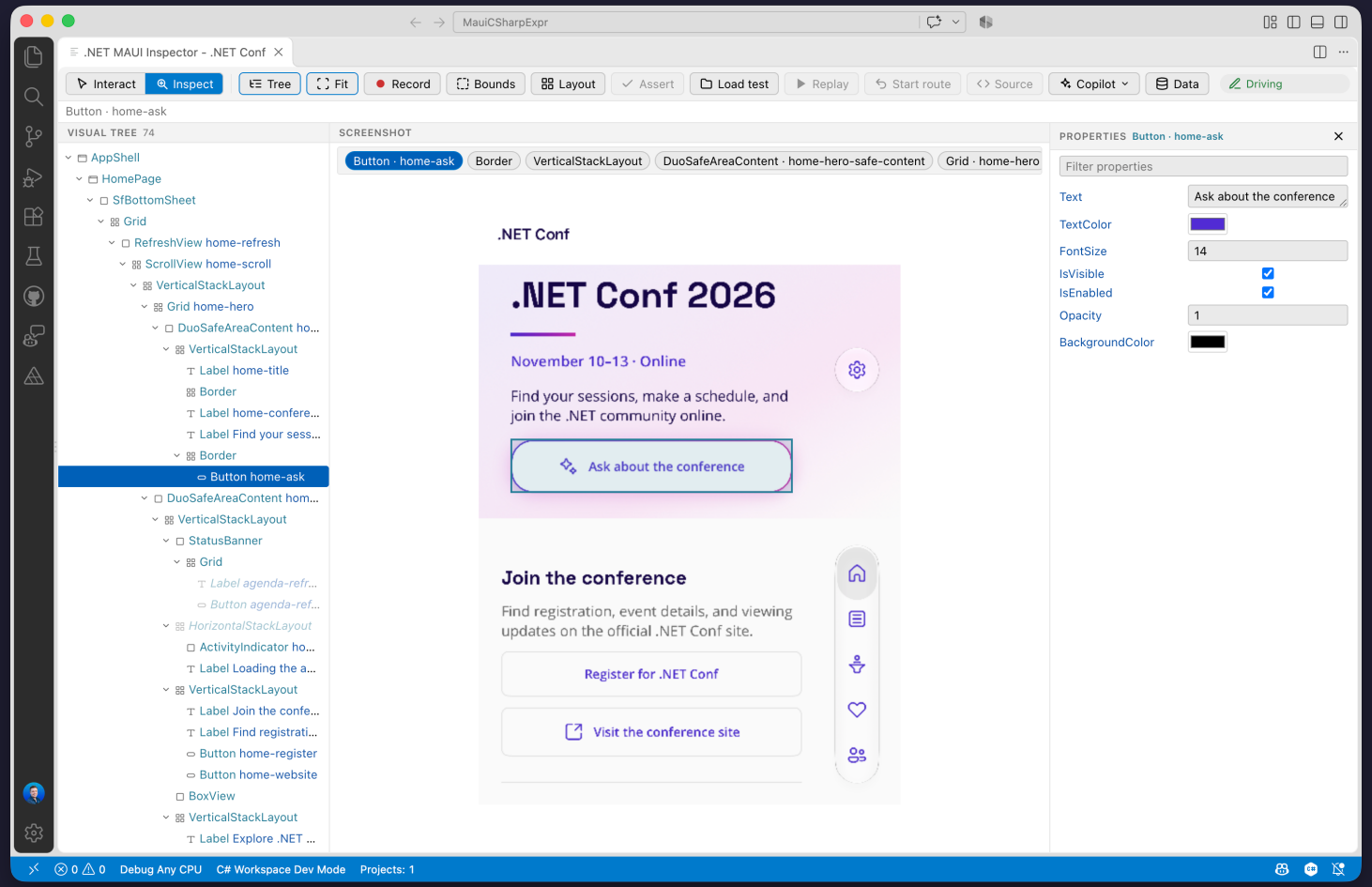
See inside your running app, and drive it

WHAT IT DOES

- Live visual tree, properties and screenshot
- Record, replay and assert UI flows
- Copilot drives the same session through MCP tools

WHERE IT RUNS

- VS Code
- GitHub Copilot app
- Copilot CLI
- Any browser



Part of MAUI Labs: github.com/dotnet/maui-labs

05 MAUI Labs & AI



Experiments to try in .NET MAUI Labs

MAUI CLI
Setup, SDKs and emulators

DevFlow
Inspect, automate and debug running apps

Essentials.AI
On-device chat and embeddings

AI Extensions
Turn methods into AI tools

Agent skills
maui ai init sets up skills and MCP

10+
experimental projects and more in the repo
Explore it all at github.com/dotnet/maui-labs

Comet & Go
MVU UI in C# and single-file prototypes

Linux backend
MAUI apps on Linux via GTK

WPF backend
22+ controls, Shell and Blazor on WPF

MAUI Testing
dotnet new mauitest for device tests

AppProjectRef
Consume .apk, .ipa and .msix in MSBuild

Four Labs tools worth trying first

MAUI CLI

One tool for setup and devices

- **maui doctor** finds and fixes environment issues
- Installs Android SDK, JDK and emulators
- Manages Xcode, simulators and runtimes
- **--json** output for CI pipelines

```
maui doctor
```

DevFlow

See and drive your running app

- Visual tree, taps and screenshots via an in-app agent
- Network monitoring and profiling
- Blazor WebViews over Chrome DevTools
- MCP server lets AI agents test your app

```
maui devflow mcp
```

Essentials.AI

AI that runs on the device

- Chat and embeddings through Microsoft.Extensions.AI
- Apple Intelligence and NaturalLanguage underneath
- Streaming, structured output and tool calling
- Image input on supported systems

```
Microsoft.Extensions.AI
```

Agent skills

Set up your AI agent in one step

- Installs MAUI and DevFlow skills
- Adds Copilot agents and MCP config
- Targets VS Code, Copilot CLI and Claude
- **--dry-run** previews every change

```
maui ai init
```

MAUI CLI

maui doctor checks your whole setup and fixes what it can.

Android SDKs, JDKs, emulators, Xcode and simulators are managed from one place.

--json and --ci make the same commands work in pipelines.

```
# Install (pre-release)
dotnet tool install -g Microsoft.Maui.Cli --prerelease

# Diagnose and auto-fix your environment
maui doctor

# Android SDK, JDK and emulators
maui android
maui android emulator

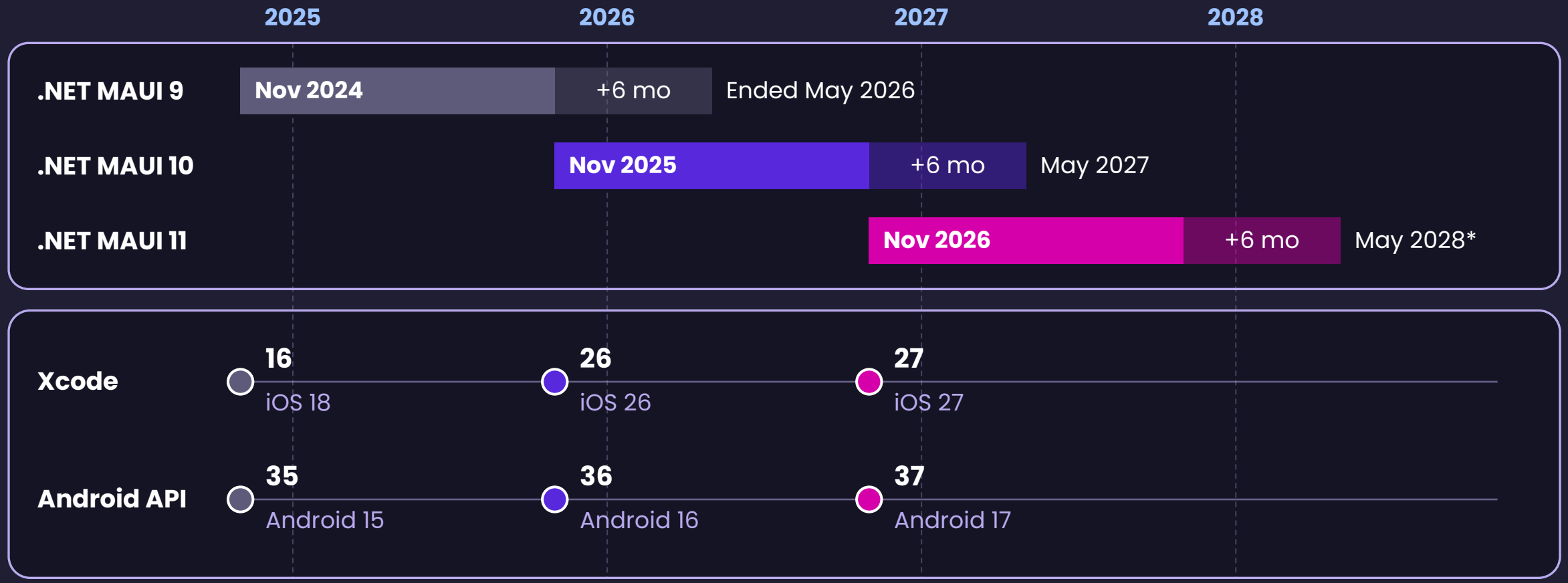
# Xcode, simulators and runtimes (macOS)
maui apple

# Every connected device, every platform
maui device list

# Machine-readable output for CI
maui doctor --json --ci
```

Each .NET MAUI version is patched for 18 months

Release schedule, support and patching



aka.ms/maui-support-policy

Support ends 6 months after the next major version ships. *.NET MAUI 11 end of support assumes .NET 12 ships November 2027.

Go deeper with .NET MAUI

Native apps for every platform, from mobile to Windows.

TUESDAY, NOVEMBER 10

BRK016
Bringing dotnet run, test, and watch to .NET MAUI
Jonathan Peppers
Tuesday • 4:30 PM PST

PRERECORDED

BRK033
Less Chat More Action: AI-Powered .NET MAUI Apps
Gerald Versluis
Prerecorded

BRK034
How We Brought CoreCLR to .NET MAUI and Mobile
Jonathan Peppers
Prerecorded

WEDNESDAY, NOVEMBER 11

BRK022
Building great Windows apps with WinUI and agents
Morten Nielsen
Wednesday • 11:30 AM PST

BRK024
CsWinRT 3.0: Modernizing the Windows Projections for .NET
Manodasan Wignarajah
Wednesday • 12:30 PM PST

BRK026
Beyond Code Generation: Give AI Eyes and Hands for .NET MAUI
Gerald Versluis
Wednesday • 1:30 PM PST

BRK027
No XAML Required: Native Windows Apps in Pure C#
Alexandre Zollinger Chohfi
Wednesday • 2:00 PM PST

Download

.NET 11

get.dot.net



Thank you

